
Chocolate Bar Company Ratings



Kaggle Dataset

Jacqueline Molina | June 2026



TABLE OF CONTENTS



01

EXECUTIVE
SUMMARY

02

OBJECTIVE

03

DATA
SOURCES

04

CLEANING &
MANIPULATION



TABLE OF CONTENTS



05

KEY FINDINGS

06

SUMMARY OF
DATA
ANALYSIS

07

NEXT STEPS

08

Q&A

EXECUTIVE SUMMARY

We analyzed and found which elements from the five major factors (*Cocoa Percentage, Company Location, Bean Origin, Broad Bean Origin, Bean Type*) directly influenced a chocolate bars' rating and cross checked these results with the top ten highly rated companies.

Key Findings

- Top performing factors when considering creating a chocolate bar
- An insight on the usage of resources from the top ten companies
- Why the worst rated companies are at the bottom

Recommended Actions

- Invest in quality products
 - Source from top locations
 - Consistency is key, but there's no harm in exploring
-

OBJECTIVES

Business Task

This dataset contains expert ratings of over 1,700 chocolate bars from different companies. Each chocolate is evaluated from a combination of objective and subjective interpretations. A rating here represents an experience with one bar from each batch. The objective is to perform exploratory data analysis and visualization to determine which company produces the best chocolate bars and what factors go into doing so.

Questions

- What are the top 10 chocolate bar companies?
 - Which countries produce the highest rated bars?
 - What factors are 'the best' when creating a chocolate bar?
 - How can we improve ratings?
-

DATA SOURCES



<https://www.kaggle.com/datasets/rtatman/chocolate-bar-ratings>

Source: Public Data from Kaggle

File Name: flavors_of_cacao.csv

Rows: 1700+ entries

Columns:

Company (Chocolate Bar manufacturer), **Specific Bean Origin** (specific geolocation of origin), **REF** (when the review was entered to the database), **Review Date** (date review was published), **Cocoa Percent** (percentage of the chocolate bar), **Company Location** (manufacturer base company), **Rating** (rating out of 5), **Bean Type** (variety of bean used), **Broad Bean Origin** (broad geolocation of origin)

CLEANING



Starting shape: (1795, 9)

- Checking all columns for missing data
- Renaming Column Names
- Dropping any null values
- Reducing Varieties of Beans from Bean Origin (1039 to 508), Broad Bean Origin (99 to 57), Bean Type (41 to 11)
- Fixing mistyped locations and company names
- Dropping Useless Columns (REF, Bean Origin, Bean Type, Broad Bean Origin) (we replaced these with their simplified versions)
- Correcting column data types to make data easier to work with (Cocoa Percentage from object to float)
- Reducing Companies by removing those that have 2 or less entries (carry the potential to skew the ratings)

Final shape: (1675, 8)

MANIPULATION



Phase 1 (Top Companies and their Usage of Resources)

- Find the top 10 chocolate bar companies by using an average overall rating of all company entries
 - Find out why these are the most highly rated companies by looking into the factors that make up chocolate bars
 - Locate where all these companies base their manufacturers
 - Find out what cocoa percentages companies are using and what rating each percentage returns
 - Compare the average rating of Company Location and Cocoa Percentage
 - Find out the type of Broad and Specific geo-specific beans the top companies use and what ratings are associated with them
 - Find out which Bean Types companies use
 - Create a DataFrame of the best performing year(s) for each top company and their average ratings
-

MANIPULATION



Phase 2 (Finding the 'Best' Objective Resources to Use)

- Top Company Locations with the highest average rating
 - Visual of distribution of chocolate companies across the world
 - Most popular used Cocoa Percentages and how many entries are found
 - Best Broad Bean Origin/Rating combination
 - Best Bean Type grouped in order of best rating found
 - Bean Origin found by taking the average of all entries with that specific bean
-

MANIPULATION



Phase 3 (Narrowing Down the 'Perfect' Chocolate Bar)

To properly rate the top ten chocolate companies, we need to know how many of them utilize the best components from each factor (Bean Type, Bean Origin, Cocoa Percentage, Broad Bean Origin, Company Location)

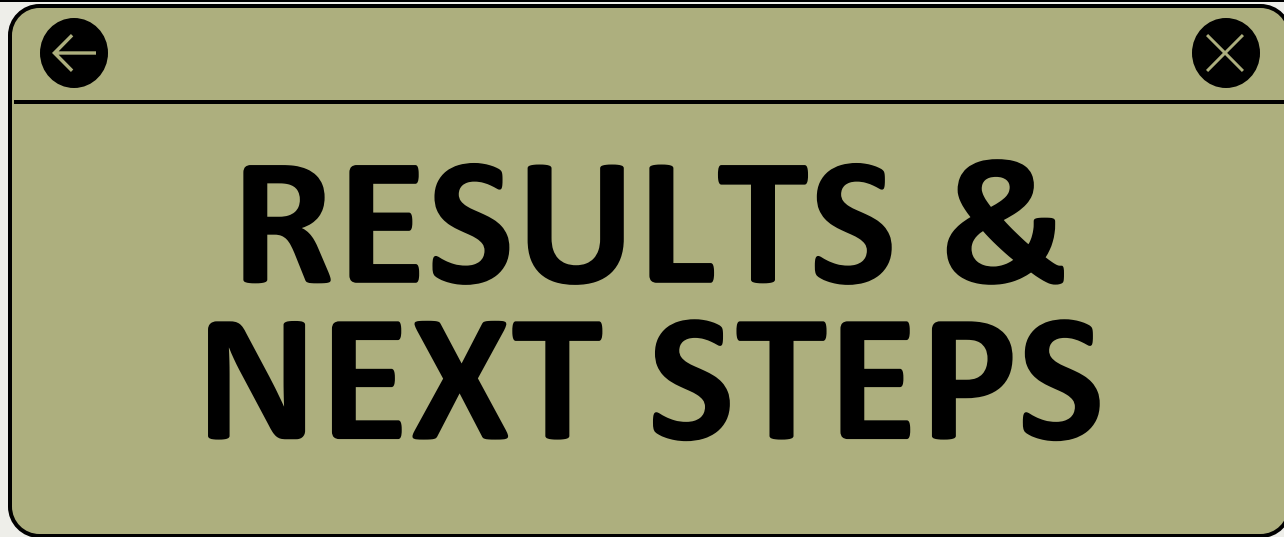
- Create subsets using the 'best' components of all factors to see which companies use them
 - Create a conditional statement that checks to see how many of the top companies use one or more top factors in their chocolate bars (starting 64 entries from top companies, all 64 entries satisfy this requirement)
 - Create a conditional statement that checks how many top companies utilize components from ALL factors in one chocolate bar (only 6 entries returned)
-

MANIPULATION



Phase 4 (Bottom 10 Companies)

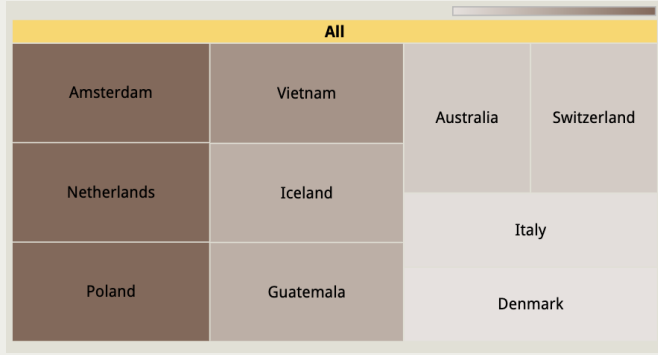
- Find the bottom 10 chocolate bar companies by using an average overall rating of all company entries
 - Create a subset of just the worst chocolate bar companies
 - Cross check which companies utilize all 'best' components from factors (none)
 - Check each factor individually to see how the bottom companies utilize their resources
-



[Tableau Public Link](#)

[Google Data Studio/Looker Link](#)

KEY FINDINGS – Objective Best



Company Locations:

Amsterdam, Netherlands, Poland, Vietnam, Iceland, Guatemala, Australia, Switzerland, Italy, and Denmark. Highest average ratings (3.35–3.5). 187 entries with companies in these locations. Italy had an outlier score of 5.0.

Top Rated (GeoSpecific) Bean Origin

	Bean Origin	Rating 
1.	Chuao	5
2.	Toscano Black	5
3.	Porcelana	4
4.	Madagascar	4
5.	Ocumare	4
6.	Piura	4

Bean Origin:

Chuao, Toscano Black, Piura, Bachelor's Hall E. Monte Alegre, Morobe, Nine, Ocular, China, and Haiti. Chuao and Toscano Black scoring a perfect 5.0 rating.

KEY FINDINGS – Objective Best



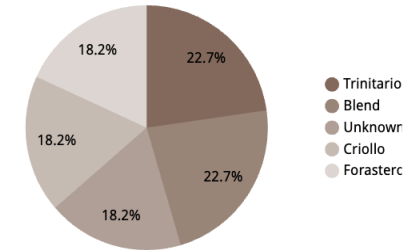
Bean Type:

Blend, Trinitario, Amazon, Criollo, Forester, Unknown, Beniano, EET, Matina, and Nacional. There are only eleven bean types, so I'm going to consider the top five.

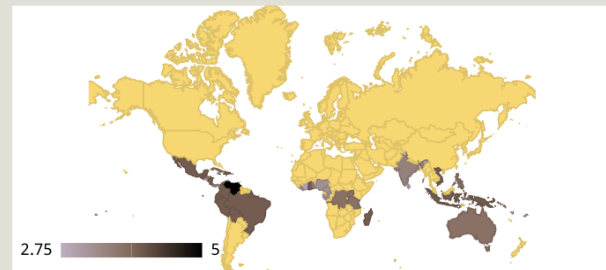
Broad Bean Origin:

Venezuela, Unknown, Vietnam, St.Lucia, South America, São Tomé, Peru, or Papua New Guinea.

Top 5 Bean Types



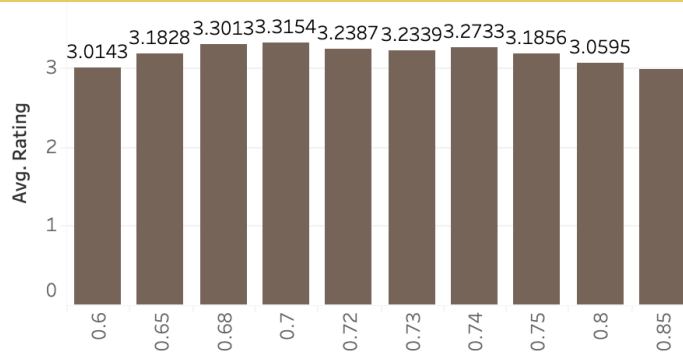
Most Popular Broad Bean Origins (Country)



KEY FINDINGS – Objective Best



Most Popular Cocoa Percentages Utilized



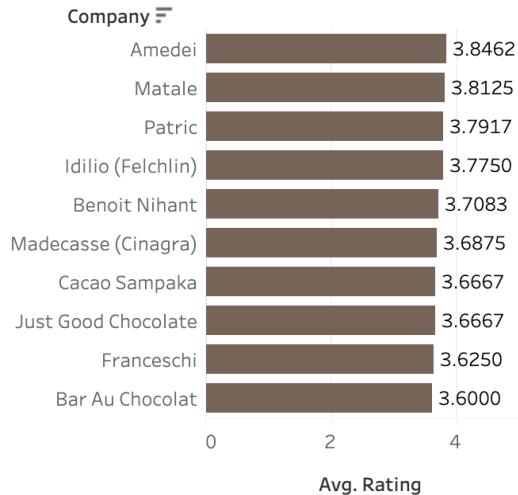
Cocoa Percentage:

Most popular percentage to use when creating chocolate bars across the globe is an extremely popular 70% with 562 entries employing this percentage. 75% (198 entries) and 72% (155 entries) are also top contenders, however noticeably lower in popularity. Still in the top 10 are 65%, 80%, 74%, 68%, 60%, 73%, and 85%. 70% on average returns a 3.5 score, while 72% returns a 3.25–3.5 score.

KEY FINDINGS – Top 10 Companies



Top 10 Chocolate Companies



Location

Three companies located in countries that were considered the best (Matale – Australia, Idilio (Felchlin) – Switzerland, and Amedei – Italy). Three other companies were located in the USA.

Broad Bean Origin

In the top ten companies data frame, 22 entries are from Venezuela with 15 being rated a 3.75 or above. Four entries are from Peru, all scoring between a 3.75–4.0. 3 from Papua New Guinea, also scoring 3.75–4.0.

Bean Type

Blend was the type that was the highest rated, but Trinitario was the type that was utilized the most within companies.

KEY FINDINGS – Top 10 Companies

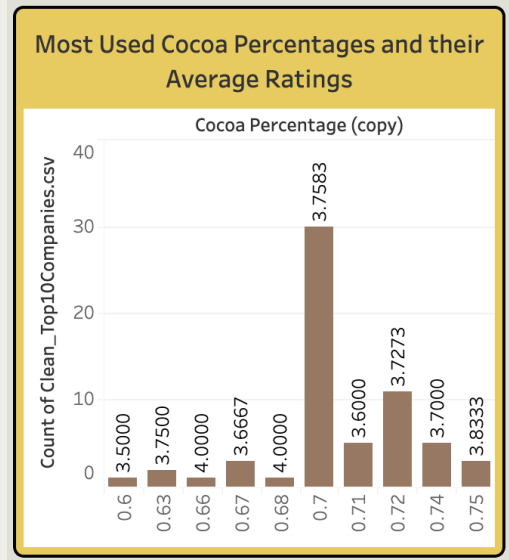


Cocoa Percentage

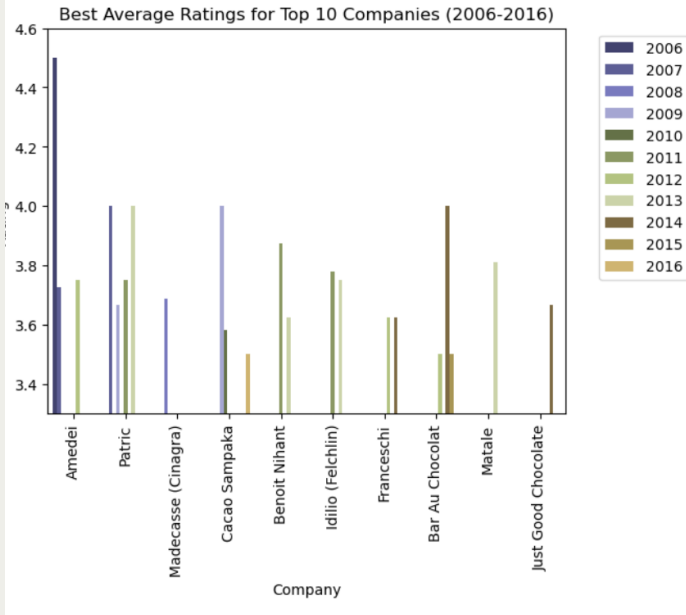
arge majority utilize the 70% (30 entries). Followed by 72% and a tie between 74% and 71%. The 70% cocoa on average returns a score of 3.5-4.0, while 72% cocoa returns average scores of 3.75-4.0.

Bean Origin

Toscano Black and Chuao remain in the top two spots.



KEY FINDINGS – Top 10 Companies



Top Performing Years For the Top 10 Companies.

- Amedei - 2006 (4.5)
- Patric - 2007/2013 (4.0)
- Madecasse (Cinagra) - 2008 (3.7)
- Cacao Sampaka - 2009 (4.0)
- Idilio (Felchlin) - 2011 (3.8)
- Benoit Nihant - 2011 (3.9)
- Franceschi - 2012 (3.7)
- Matale - 2013 (3.8)
- Just Good Chocolate - 2014 (3.7)
- Bar Au Chocolat - 2014 (4.0)



SUMMARY OF DATA ANALYTICS (FINDINGS)

We analyzed the five major factors (Cocoa Percentage, Company Location, Bean Origin, Broad Bean Origin, Bean Type) that contribute to a chocolate bars' rating and cross checked these results with the top ten companies.

Amedei Chocolate Company, Idilio (Felchlin), Matale, Patric, and Benoit Nihant are the ideal companies. They utilize resources from all components that are considered to be the 'best'.

When including those entries that met all criteria for higher ratings, only six were returned. From those six entries, two scored a perfect 5.0.

The bottom ten companies all scored between 2.5 - 2.7. None met the criteria of fulfilling all 'best' factors in their chocolate bars. None were from desirable locations. These companies did not employ these components in the right combinations.



SUMMARY OF DATA ANALYTICS (IMPLICATIONS)

The companies that decide to factor in more components from the best features are likely going to be the most successful in the long term.

Location seems to be the most important factor in the manufacturing process, with cocoa percentage being the least.

The top ten companies are at the top because they employ the use of the best resources possible.

If more companies used different Broad Bean Origins (located in the best top ten) and Bean Types (Best top five), they would receive better ratings in the long run.

NEXT STEPS



Company Location

3/10 top companies were in desired locations, while 0/10 bottom companies found. All companies should take advantage of the 'best' locations to manufacture chocolate and allocate resources there.

Underutilized Resources

All companies should utilize the lesser used 'best' resources. Amazon, Criollo are hardly used Bean Types. Vietnam and Sao Tome are underutilized Broad Bean Origins. Etc.

Improving Ratings

Companies begin to exclusively choose from the objective best resources to maximize ratings. For those tight on resources, choose from the most critical factors (Location (3/10 top), Bean Origin(5/10 top))

THANKS!

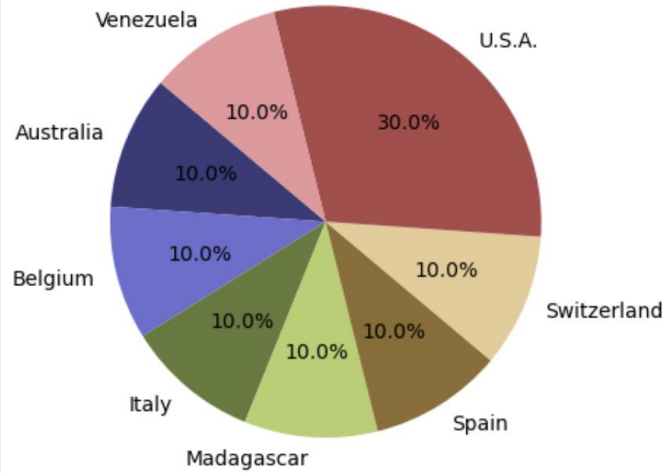
Any Questions?

CREDITS: This presentation template was created
by *Slidesgo*, including icons by *Flaticon*,
infographics & images by *Freepik*.

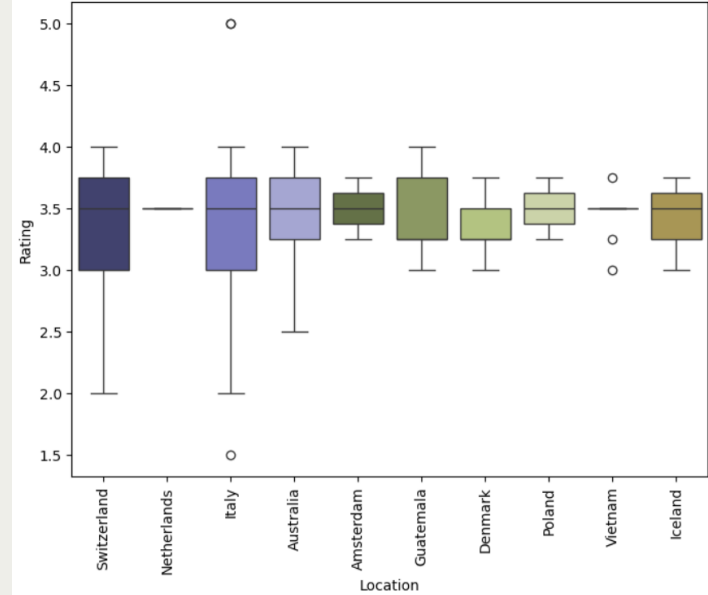


APPENDIX

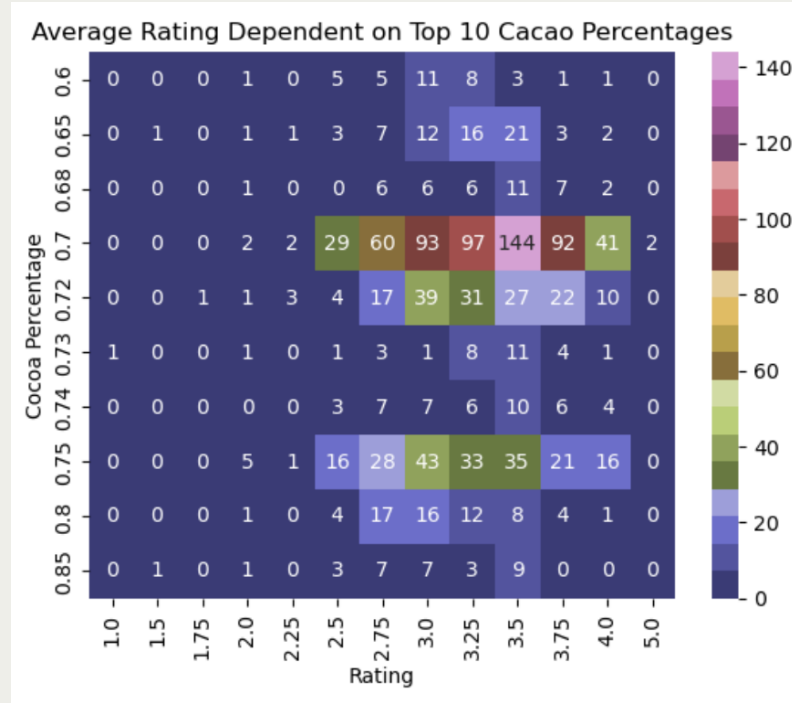
Distribution of Top 10 Company Locations



Box Plot of Highest Ratings for Top 10 Company Locations



APPENDIX



APPENDIX

Distribution of Top 10 Companies Broad Bean Origin Usage

